

SCHOOL OF TECHNOLOGY



DATABASE MANAGEMENT SYSTEMS
PROJECT-BASED LEARNING (PBL) REPORT

Small Database Application Development using Python

PROJECT TITLE

**DESIGN AND IMPLEMENTATION OF A DATABASE MANAGEMENT SYSTEM FOR INTERNSHIP
APPLICATION AND PROGRESS MONITORING SYSTEM**

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This PBL activity has strengthened my understanding of DBMS concepts and provided valuable experience in applying them to a real-world problem.

UNDERTAKING

I hereby declare that the project entitled **Design and Implementation of a database management sysem for intership application and progress monitoring system** ,submitted as part of the Project-Based Learning activity for the **Database Management Systems (DBMS)** course, is my individual work carried out under the guidance of the course faculty.

I undertake that the project report, database design, SQL scripts, application source code, results, and supporting materials submitted by me have been prepared for academic purposes. Wherever information, ideas, tools, datasets, or other resources have been referred to, appropriate acknowledgement has been provided.

I further undertake that I am responsible for the correctness and authenticity of the work submitted and that I have followed the academic and project guidelines prescribed by the University.

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LIST OF ACRONYMS

Acronym	Expansion
DBMS	Database Management System
PBL	Project-Based Learning
IAPMS	Internship Application and Progress Monitoring System
SQL	Structured Query Language
ERD	Entity-Relationship Diagram
CRUD	Create, Read, Update and Delete
3NF	Third Normal Form
DDL	Data Definition Language
PK	Primary Key
FK	Foreign Key
HTML	Hypertext Markup Language
CSS	Cascading Style Sheets
JS	JavaScript

Abstract:

The Internship Application and Progress Monitoring System (IAPMS) is a web-based database management system that will be used to manage the life-cycle of internships for students in an organized and centralized manner. The system solves the problems of maintaining internship opportunities, student applications, approvals, the assignment of mentors and supervisors from companies, weekly progress reporting, evaluation, certificate generation and feedback through independent or manual records. MySQL relational database with 12 entities and their correct primary keys, foreign keys, unique constraints, NOT NULL constraints and CHECK constraints and default values to ensure data consistency and integrity. The user interface and the links between the application layer and the database are provided by a Python Flask application. The system enables students to enroll in internships, log in, view internships in which they are enrolled for weekly progress monitoring, submit application, monitor applications, enter evaluation/ certificate data, and manage internships. Data is accessed via SQL queries, SQL joins, SQL views, and SQL aggregate operations, to generate meaningful data and reports. It includes validation, error handling, CRUD operations, and functional testing, ensuring system reliability. The proposed system will provide a structured solution to maintaining internship related information and so it will make the internship related records easily accessible, consistent and organized.

Keywords:

- Internship Management
- Database Management System
- MySQL
- Python Flask, Progress Monitoring
- Student Applications

CHAPTER-1

PROBLEM STATEMENT, SOLUTION STRATEGY AND ER DIAGRAM

1.1 Problem Statement:

Internship management process consists of the activities like keeping track of internship opportunities, receiving internship applications, processing, assigning faculty mentors and supervisor of the company, monitoring internship progress weekly, evaluating, giving internship certificate, and taking feedback from the interns. If these activities are not kept in separate files, spreadsheets or manual records, it's hard to manage and retrieve internship-related information.

Without a centralized system, this could lead to duplicate data, inconsistent data, hard to track status of applications, etc., and hard to track students' internship progress. It can also be challenging for administrators and faculty members to get relevant information needed and create reports when necessary.

The Internship Application and Progress Monitoring System (IAPMS) has been designed to solve these problems by using a web-based system implemented in a central database. The system provides a MySQL relational database to organize internship related information, and offers a Python Flask based application to interact with the database.

The proposed system takes care of the major phases of the internship lifecycle such as internship opportunity management, student login and registration, submission of internship applications and tracking, internship application approval, assigning mentors and supervisors, progress monitoring every week, evaluation, internship certificate information, and feedback. To ensure data integrity and reliable system functioning, database constraints, validation, CRUD operations, SQL queries, and functional testing are integrated.

1.2 Objectives:

The main objective of the Internship Application and Progress Monitoring System (IAPMS) is to develop a centralized database-driven web application that provides an organized and reliable approach for managing the internship lifecycle of students. The specific objectives of the system are:

1. To develop a centralized database for storing and managing internship-related information.
2. To maintain student and organization information, including student details, departments, internship providers, and contact information.
3. To manage internship opportunities by storing details such as internship title, description, location, duration, and available slots.
4. To enable students to register and authenticate securely before accessing student-specific services.
5. To provide internship application management, allowing students to apply for available opportunities and track their application status.
6. To manage internship approvals and assignments by associating approved applications with faculty mentors and company supervisors.
7. To monitor weekly internship progress by maintaining student activity logs, dates, working hours, and submission status.
8. To maintain evaluation, certificate, and feedback information associated with completed or ongoing internships.
9. To implement CRUD operations and SQL-based reporting for efficient management and retrieval of internship information.
10. To maintain data integrity and reliability using database constraints, application-level validation, error handling, and functional testing.

1.3 Scope and Assumptions:

Scope:

The scope of the **Internship Application and Progress Monitoring System (IAPMS)** covers the major activities required to manage student internships through a centralized database-driven web application. The system includes:

- Student registration, login, and student information management.
- Organization and internship opportunity management.
- Internship application submission and status tracking.
- Application approval and internship assignment.
- Faculty mentor and company supervisor management.
- Weekly internship progress monitoring.
- Evaluation, certificate, and feedback management.
- CRUD operations and generation of reports using SQL queries, joins, views, and aggregate operations.

Assumptions:

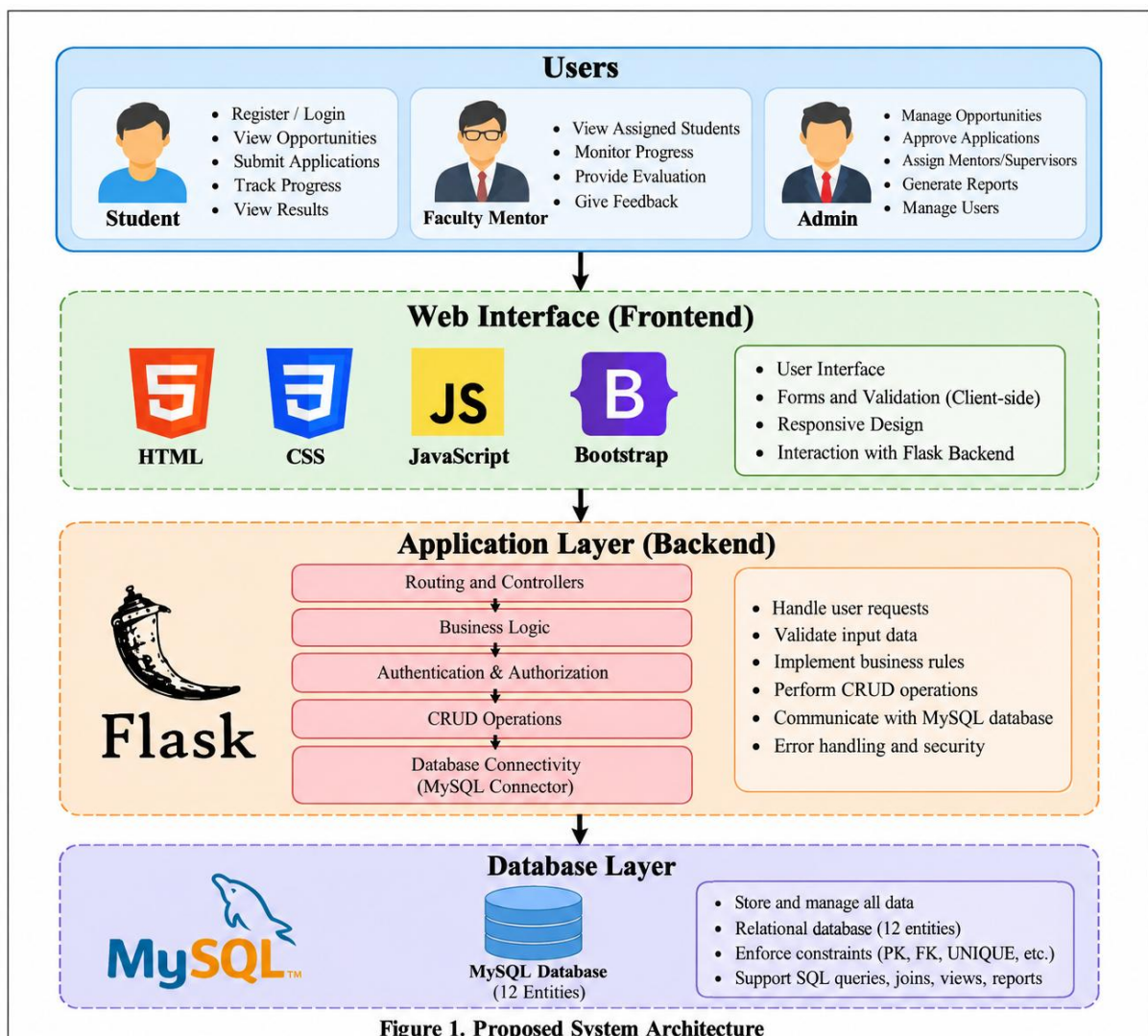
- The system is developed based on the following assumptions:
- Each student has a unique student ID and email address.
- Each internship opportunity is associated with an organization.
- A student can submit only one application for a particular opportunity.
- An approved application can be assigned to a faculty mentor and company supervisor.
- Weekly progress, evaluation, certificate, and feedback records are associated with the internship assignment.
- Users provide valid information while using the system.
- The system uses **MySQL** for database management and **Python Flask** for application development.

1.4 Proposed Solution Strategy:

The proposed Internship Application and Progress Monitoring System (IAPMS) is based on a centralized relational database and a web-based application to handle activities related to internships. The system is developed with the python flask application framework and in the database management system, the MySQL database is used.

The solution is structured in a way that allows the students to register to the web application, log in, browse for available internships and apply for them. Application information is stored in MySQL database which can be accessed for tracking & management. Approved applications will have faculty mentors and company supervisors linked in to monitor internship accounts.

In the internship, weekly progress records are kept, evaluated, and information on the internship certificate and feedback are stored in the database. Internship opportunities can be created, viewed, updated or deleted with administrative operations. Information is retrieved, and reports are created, using SQL queries, joins, views and aggregate functions.



1.5 Functional Requirements:

The functional requirements define the major operations that the Internship Application and Progress Monitoring System (IAPMS) must provide to its users. The requirements cover student activities, internship management, progress monitoring, evaluation, and reporting.

Table 1: Functional Requirements

FR ID	Functional Requirement
FR1	Student registration and login
FR2	View available internship opportunities
FR3	Submit and track internship applications
FR4	Manage application approvals
FR5	Assign faculty mentors and company supervisors
FR6	Record and monitor weekly internship progress
FR7	Maintain internship evaluations
FR8	Maintain certificate information
FR9	Record internship feedback
FR10	Create, view, update, and delete internship opportunities
FR11	Generate information using SQL queries, joins, views, and aggregate operations

System Users:

- **Student:** Registers, logs in, views opportunities, submits applications, tracks applications, and records progress.
- **Administrator:** Manages internship opportunities and related administrative operations.
- **Faculty Mentor / Company Supervisor:** Their details are maintained for internship assignment and evaluation.

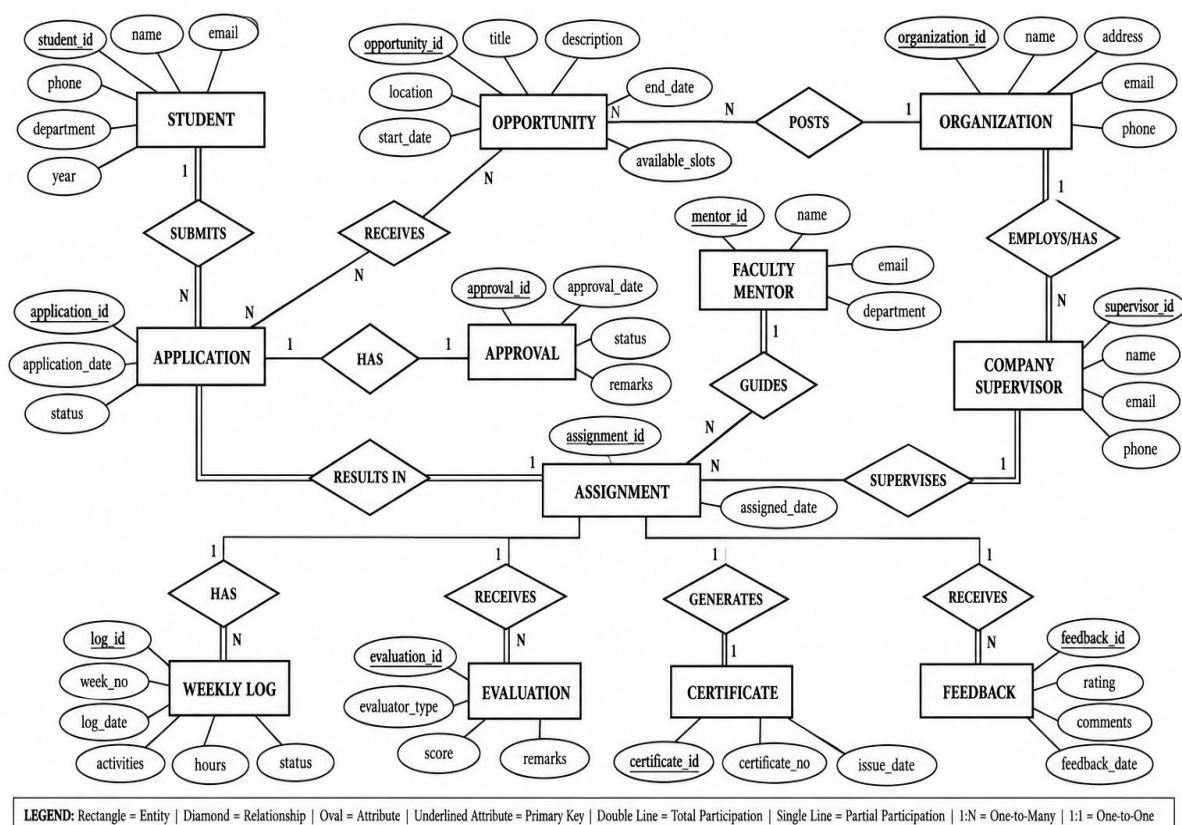
1.6 Entity-Relationship Diagram:

The Entity Relationship Diagram (ERD) depicts the conceptual structure of the Internship Application and Progress Monitoring System (IAPMS). It explains the main entities involved in the internship management process, their attributes, relationships and cardinalities. The ER diagram is the basis of transformation of the conceptual design of the database to the final relational schema.

The system has 12 main entities such as Student, Organization, Opportunity, Application, Approval, Faculty Mentor, Company Supervisor, Assignment, Weekly Log, Evaluation, Certificate and Feedback.

These entities and relationships show the flow of internship information from the creation of internship opportunities and student applications, to approval, assignment, monitoring progress, evaluation, certification, and feedback.

Figure 2. Final Entity-Relationship Diagram



1.7 Entity and Relationship Summary:

The ER diagram consists of 12 entities that represent the major components of the internship management system. These entities store student, organization, application, assignment, progress, evaluation, certificate, and feedback information.

Table 2. Entity Summary

Entity	Purpose
Student	Stores student personal and academic information.
Organization	Stores details of organizations offering internships.
Opportunity	Stores available internship opportunity details.
Application	Records student applications for internship opportunities.
Approval	Maintains approval status and remarks for applications.
Faculty Mentor	Stores faculty mentor information for internship guidance.
Company Supervisor	Stores information about supervisors from organizations.
Assignment	Records the assignment of an approved internship to mentors and supervisors.
Weekly Log	Stores weekly internship activities, hours, and submission status.
Evaluation	Maintains internship evaluation scores and remarks.
Certificate	Stores internship certificate details and issue information.
Feedback	Records internship ratings, comments, and feedback dates.

The Entity Relationship Diagram (ERD) describes the conceptual design of the Internship Application and Progress Monitoring System (IAPMS). It contains the main entities participating in the intern management process, their attributes, relationships and cardinalities. The ER diagram is the basis of transformation of the conceptual design of database to the final relational schema.

Chapter 2: LOGICAL DATABASE DESIGN

2. Logical Database Design:

The process of converting a conceptual design, such as an ER diagram, into a structured relational database model is called logical database design. The design defines the relations, attributes, primary keys, foreign keys and integrity constraints that are needed to store consistently the information related to the internship. The database has been implemented in MySQL and it contains 12 related relations which correspond to the identified entities in the ER diagram.

Table 3: Relational Schema

Relation	Attributes
STUDENT	student_id, name, email, phone, department, year
ORGANIZATION	organization_id, name, address, email, phone
OPPORTUNITY	opportunity_id, organization_id, title, description, location, start_date, end_date, available_slots
APPLICATION	application_id, student_id, opportunity_id, application_date, status
APPROVAL	approval_id, application_id, approval_date, status, remarks
FACULTY_MENTOR	mentor_id, name, email, department
COMPANY_SUPERVISOR	supervisor_id, organization_id, name, email, phone
ASSIGNMENT	assignment_id, application_id, mentor_id, supervisor_id, assigned_date
WEEKLY_LOG	log_id, assignment_id, week_no, log_date, activities, hours, status
EVALUATION	evaluation_id, assignment_id, evaluator_type, score, remarks
CERTIFICATE	certificate_id, assignment_id, certificate_no, issue_date
FEEDBACK	feedback_id, assignment_id, rating, comments, feedback_date

2.1 Keys and Integrity Constraints:

A functional dependency (FD) is a relation between the values of an attribute or group of attributes and the values of another attribute or group of attributes.

Table 4. Important Functional Dependencies

Relation	Functional Dependency
STUDENT	student_id $\rightarrow$ name, email, phone, department, year
ORGANIZATION	organization_id $\rightarrow$ name, address, email, phone
OPPORTUNITY	opportunity_id $\rightarrow$ organization_id, title, description, location, start_date, end_date, available_slots
APPLICATION	application_id $\rightarrow$ student_id, opportunity_id, application_date, status
APPROVAL	approval_id $\rightarrow$ application_id, approval_date, status, remarks
FACULTY_MENTOR	mentor_id $\rightarrow$ name, email, department
COMPANY_SUPERVISOR	supervisor_id $\rightarrow$ organization_id, name, email, phone
ASSIGNMENT	assignment_id $\rightarrow$ application_id, mentor_id, supervisor_id, assigned_date
WEEKLY_LOG	log_id $\rightarrow$ assignment_id, week_no, log_date, activities, hours, status
EVALUATION	evaluation_id $\rightarrow$ assignment_id, evaluator_type, score, remarks
CERTIFICATE	certificate_id $\rightarrow$ assignment_id, certificate_no, issue_date
FEEDBACK	feedback_id $\rightarrow$ assignment_id, rating, comments, feedback_date

These dependencies ensure that the primary key of the relation will uniquely identify the non-key attributes of the relation. There are further candidate key dependencies resulting from the UNIQUE constraints, for example (student_id, opportunity_id) determines an application record and application_id uniquely determines its approval and assignment records.

2.3 Normalization up to 3NF:

Normalization is the technique used to structure data in a systematic manner so that data redundancy can be minimized and the problems of insertion, deletion, and update anomalies can be avoided. The Internship Application and Progress Tracking System meets the conditions for Third Normal Form (3NF).

First Normal Form (1NF)

A relation satisfies 1NF when all attributes have atomic values and there are no repeating groups. In the internship database, there is only one value in every field. For instance, one STUDENT has one email, one department, and one year. The weekly activities of the intern are recorded in separate instances in the WEEKLY_LOG.

Second Normal Form (2NF)

A relation satisfies 2NF when it is in 1NF and every non-key attribute is functionally dependent on the entire candidate key. Take an example of an application which has a candidate key made up of (student_id, opportunity_id). Details of the student depend on student_id alone whereas details of the opportunity depend on opportunity_id alone. Hence, the student and opportunity details are separated from the relation.

Third Normal Form (3NF)

The condition of 3NF is that the relation must be in 2NF, and there should not be any transitive dependency of non-key attributes on the primary key. For instance, the organization details are kept in ORGANIZATION and not OPPORTUNITY. Likewise, mentor, supervisor, evaluation, certification, and feedback details are kept in different relations.

Normalization Result

Normal Form	Application in IAPMS
1NF	All attributes contain atomic values and repeating data is separated.
2NF	Partial dependencies are removed by separating student and opportunity information from application data.
3NF	Transitive dependencies are removed by separating organizations, mentors, supervisors, and other independent information.

2.4 Data Dictionary — Part 1:

The data dictionary describes the attributes of the database relations, including their data types, keys, constraints, and purpose. It provides a detailed reference for the logical database design.

Table 5. Data Dictionary — Part 1

Table	Attribute	Data Type	Key / Constraint	Description
STUDENT	student_id	INT	PK, AUTO_INCREMENT	Unique student identifier
	name	VARCHAR(100)	NOT NULL	Student name
	email	VARCHAR(100)	UNIQUE, NOT NULL	Student email
	phone	VARCHAR(15)	—	Contact number
	department	VARCHAR(100)	—	Academic department
	year	INT	NOT NULL, CHECK 1–4	Current academic year
	password_hash	VARCHAR(255)	—	Hashed login password
ORGANIZATION	organization_id	INT	PK, AUTO_INCREMENT	Unique organization identifier
	name	VARCHAR(150)	NOT NULL	Organization name
	address	VARCHAR(255)	—	Organization address
	email	VARCHAR(100)	UNIQUE	Organization email
	phone	VARCHAR(15)	—	Contact number
OPPORTUNITY	opportunity_id	INT	PK, AUTO_INCREMENT	Unique opportunity identifier
	organization_id	INT	FK, NOT NULL	Associated organization
	title	VARCHAR(150)	NOT NULL	Internship title
	description	TEXT	—	Internship description
	location	VARCHAR(150)	—	Internship location
	start_date	DATE	NOT NULL	Internship start date
	end_date	DATE	NOT NULL	Internship end date
	available_slots	INT	NOT NULL, CHECK > 0	Available internship slots

2.4 Data Dictionary — Part 2:

Table 6. Data Dictionary — Part 2:

Table	Attribute	Data Type	Key / Constraint	Description
APPLICATION	application_id	INT	PK, AUTO_INCREMENT	Unique application identifier
	student_id	INT	FK, NOT NULL	Applicant student
	opportunity_id	INT	FK, NOT NULL	Applied internship
	application_date	DATE	NOT NULL	Application date
	status	VARCHAR(20)	NOT NULL, CHECK	Application status
APPROVAL	approval_id	INT	PK, AUTO_INCREMENT	Unique approval identifier
	application_id	INT	FK, UNIQUE	Related application
	approval_date	DATE	NOT NULL	Decision date
	status	VARCHAR(20)	CHECK	Approval status
	remarks	VARCHAR(255)	—	Approval remarks
FACULTY_MENTOR	mentor_id	INT	PK, AUTO_INCREMENT	Unique mentor identifier
	name	VARCHAR(100)	NOT NULL	Mentor name
	email	VARCHAR(100)	UNIQUE, NOT NULL	Mentor email
	department	VARCHAR(100)	NOT NULL	Mentor department
COMPANY_SUPERVISOR	supervisor_id	INT	PK, AUTO_INCREMENT	Unique supervisor identifier
	organization_id	INT	FK, NOT NULL	Associated organization
	name	VARCHAR(100)	NOT NULL	Supervisor name
	email	VARCHAR(100)	UNIQUE	Supervisor email
	phone	VARCHAR(15)	—	Contact number

Table 7. Data Dictionary — Part 2 (Continued):

Table	Attribute	Data Type	Key / Constraint	Description
ASSIGNMENT	assignment_id	INT	PK, AUTO_INCREMENT	Assignment identifier
	application_id	INT	FK, UNIQUE, NOT NULL	Approved application
	mentor_id	INT	FK, NOT NULL	Faculty mentor
	supervisor_id	INT	FK, NOT NULL	Company supervisor
	assigned_date	DATE	NOT NULL	Assignment date
WEEKLY_LOG	log_id	INT	PK, AUTO_INCREMENT	Progress-log identifier
	assignment_id	INT	FK, NOT NULL	Related assignment
	week_no	INT	NOT NULL, CHECK > 0	Week number
	log_date	DATE	NOT NULL	Log date
	activities	TEXT	NOT NULL	Activities performed
	hours	DECIMAL(5,2)	NOT NULL, CHECK ≥ 0	Hours completed
EVALUATION	evaluation_id	INT	PK, AUTO_INCREMENT	Evaluation identifier
	assignment_id	INT	FK, NOT NULL	Related assignment
	evaluator_type	VARCHAR(20)	CHECK	Faculty or Company
	score	DECIMAL(5,2)	CHECK 0–100	Evaluation score
CERTIFICATE	certificate_id	INT	PK, AUTO_INCREMENT	Certificate identifier
	assignment_id	INT	FK, UNIQUE	Related assignment
	certificate_no	VARCHAR(50)	UNIQUE, NOT NULL	Certificate number
	issue_date	DATE	NOT NULL	Certificate issue date
FEEDBACK	feedback_id	INT	PK, AUTO_INCREMENT	Feedback identifier
	assignment_id	INT	FK, NOT NULL	Related assignment
	rating	INT	CHECK 1–5	Feedback rating
	comments	VARCHAR(500)	—	Feedback comments
	feedback_date	DATE	NOT NULL	Feedback date

2.5 DDL and Sample Data:

Database Schema is created using DDL statements in MySQL. DDL statements are used to create tables define attributes set up keys and foreign keys and add integrity constraints. Some examples of DDL statements are listed below. The full database code can be found in Appendix- A.

Representative DDL

```
CREATE TABLE student (  
    student_id INT PRIMARY KEY AUTO_INCREMENT,  
    name VARCHAR(100) NOT NULL,  
    email VARCHAR(100) NOT NULL UNIQUE,  
    phone VARCHAR(15),  
    department VARCHAR(100),  
    year INT NOT NULL,  
    password_hash VARCHAR(255),  
    CHECK (year BETWEEN 1 AND 4)  
);  
  
CREATE TABLE application (  
    application_id INT PRIMARY KEY AUTO_INCREMENT,  
    student_id INT NOT NULL,  
    opportunity_id INT NOT NULL,  
    application_date DATE NOT NULL,  
    status VARCHAR(20) NOT NULL DEFAULT 'Applied',  
    UNIQUE (student_id, opportunity_id),
```

```

CHECK (status IN

('Applied','Shortlisted','Rejected','Approved')),

FOREIGN KEY (student_id)

REFERENCES student(student_id),

FOREIGN KEY (opportunity_id)

REFERENCES opportunity(opportunity_id)

);

```

Sample Data:

Table	Sample Records
STUDENT	Rahul Sharma, Priya Reddy, Arjun Kumar
ORGANIZATION	TechNova Solutions, DataSphere Analytics
OPPORTUNITY	Python Developer Intern, Data Analyst Intern
APPLICATION	Student applications with Applied/Approved status
FACULTY_MENTOR	Assigned faculty mentor records
COMPANY_SUPERVISOR	Company supervisor records

The database has example records in the tables to show how data gets added, how things are connected, how to ask questions and how the app works. The main keys and the keys that link tables help keep the data correct and make sure everything matches up. Rules stop data from being added. All the SQL code for making the database setting up rules adding data and creating views is, in Appendix A.

Chapter 3: APPLICATION DEVELOPEMENT

3.1 Application Overview

The Internship Application and Progress Monitoring System (IAPMS) is built as a Python Flask web application that connects to a MySQL database. This system gives students a way to access internship services. Provides administrators with tools to manage internship opportunities.

The main features of the system include:

- Students can. Log in to the system.
- They can view internship opportunities that're currently available.
- They can apply for internships. Keep track of their applications.
- They can check the status of their submitted applications.
- They can. Monitor their weekly internship progress.
- They can view their evaluation results, certificates and feedback
- Administrators can manage internship opportunities, such, as adding or updating them.
- The system uses the database to retrieve information and generate reports.

The system is designed with three layers:

The Presentation Layer uses HTML and CSS to create web pages. Flask templates help display the user interface.

The Application Layer is built with Python Flask. It handles user requests manages sessions checks data runs business logic and connects to the database.

The Database Layer uses MySQL to store and organize all internship data.

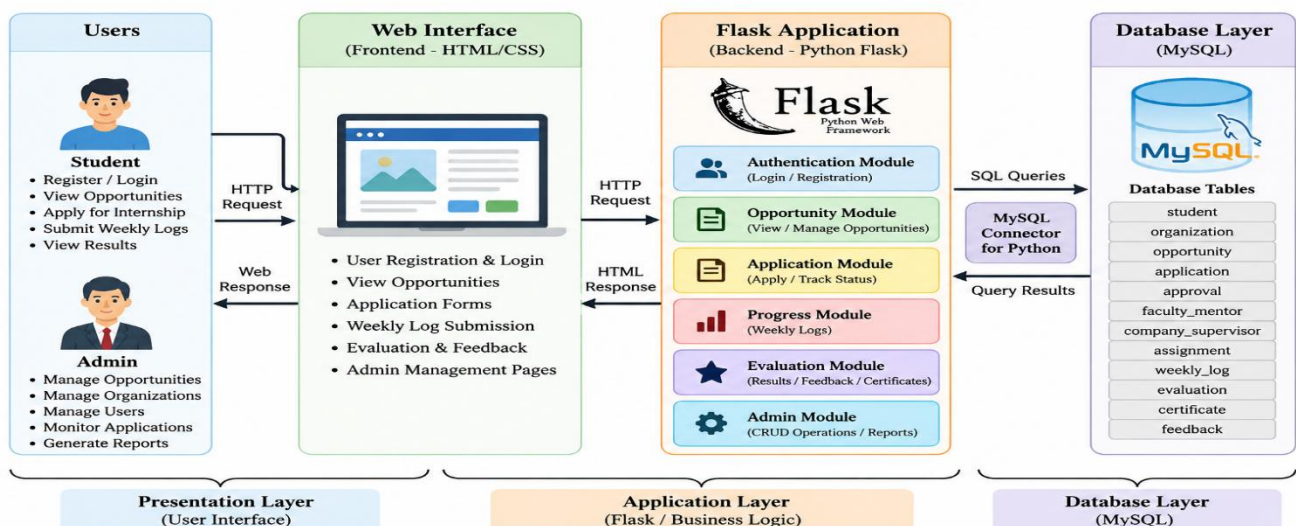


Figure 3. Architecture of the Internship Application and Progress Monitoring System

3.2 Database Connectivity and Module-to-Database Mapping:

Development Environment and Database Connectivity

The application is written in Python using Flask and MySQL. Flask processes web requests, defines routing within the application, manages sessions, does validation, and communicates with the database. The application communicates with MySQL through MySQL Connector for Python.

The connection to the database is made in `database.py` in the `get_db_connection()` function. All modules of the application use this connection to run SQL queries and manipulate database entries.

Database Connectivity

```
import mysql.connector

def get_db_connection():

    connection = mysql.connector.connect(

        host="localhost",

        user="root",

        password="*****",

        database="internship_management"

    )

    return connection
```

The connection is established when database operations are required and is closed after the operation is completed. This approach allows the Flask application to communicate with the MySQL database while maintaining a structured separation between application logic and database operations.

3.3 Application Module-to-Database Mapping

Table 8. Application Module-to-Database Mapping

Application Module	Main Database Tables
Registration & Login	STUDENT
Student Dashboard	STUDENT, APPLICATION
Opportunities	ORGANIZATION, OPPORTUNITY
Applications	STUDENT, OPPORTUNITY, APPLICATION
Approval & Assignment	APPLICATION, APPROVAL, ASSIGNMENT
Progress Monitoring	ASSIGNMENT, WEEKLY_LOG
Results	ASSIGNMENT, EVALUATION, CERTIFICATE, FEEDBACK
Admin Management	ORGANIZATION, OPPORTUNITY
Reports	Multiple related tables and database views

The module-to-table mapping ensures that each application function operates on the appropriate database relations and maintains consistency between the web application and the underlying database.

3.4 CRUD Operations:

CRUD operations represent the four basic database operations: Create, Read, Update, and Delete. In the IAPMS application, CRUD functionality is implemented for managing internship opportunities through the administrative interface.

Table 9. CRUD Operations in the Application

Operation	Application Function	Database Operation
Create	Add a new internship opportunity	INSERT
Read	View available internship opportunities	SELECT
Update	Edit internship details	UPDATE
Delete	Remove an internship opportunity	DELETE

Create

Administrator can create a new internship opportunity by providing details like organization, name of the opportunity, description of the opportunity, its location, dates, and number of spots available. These details are then validated and inserted into the OPPORTUNITY table in the database.

Read

Internship opportunities are read from the database and displayed to the user on the screen. The details about organization are joined with the opportunity records whenever required.

Update

Administrator can update existing details of the internship opportunities like name of the opportunity, its description, location, and number of spots available. This operation is done within the MySQL database.

Delete

Administrator can delete an internship opportunity when it is no longer required. Corresponding record deletion operation is carried out by the application.

CRUD Workflow:-

Admin → Admin Dashboard → Select Opportunity → Create / Read / Update / Delete → MySQL Database → Updated Information

CRUD operation provides the facility of direct interaction between the application interface and database.

3.5 Business Rules, Validation and Error Handling:

The application applies business rules and validation to maintain data accuracy and database integrity. Validation is supported by both Flask application logic and MySQL constraints.

Table 10. Business Rules and Validation:

Area	Rule / Validation
Student	Email must be unique; year must be 1–4.
Opportunity	End date $\geq$ start date; available slots > 0 .
Application	One application per student and opportunity.
Approval	One approval record per application.
Assignment	One assignment per application.
Weekly Log	Week number > 0 ; hours ≥ 0 .
Evaluation	Score must be 0–100; evaluator must be Faculty or Company.
Certificate	Certificate number must be unique.
Feedback	Rating must be between 1 and 5.

Validation and Error Handling:

The application validates user input before database operations. Duplicate registrations and invalid login credentials are handled with appropriate error messages. Database constraints prevent invalid or inconsistent records.

3.6 Reports and Database Operations:

The system uses SQL queries, joins, aggregate functions, and database views to retrieve meaningful internship information and generate reports.

Major Database Operations:

Operation	Purpose
Student Applications	Displays applications submitted by students.
Approved Internships	Retrieves approved internship applications.
Assignment Report	Displays student, organization, internship, mentor, and supervisor details.
Weekly Progress	Retrieves internship progress records.
Evaluation Report	Displays evaluation scores and remarks.
Certificate Report	Displays certificate information.
Organization Summary	Counts applications received by each organization.

Database View:

The system includes the `internship_assignment_report` view, which combines data from student, application, opportunity, organization, faculty mentor, company supervisor, and assignment tables.

```
SELECT *  
  
FROM internship_assignment_report;
```

The view makes it easier to get information about assignments. Helps create reports without having to type the full join query over and over again.

These database tasks show how to use queries to get and display information, about internships.

3.7 Application Screens:

The developed web application provides a simple interface for students to access internship-related functions. The major student-facing screens are presented below.

Student Login:

The login screen allows registered students to enter their email and password to access the application

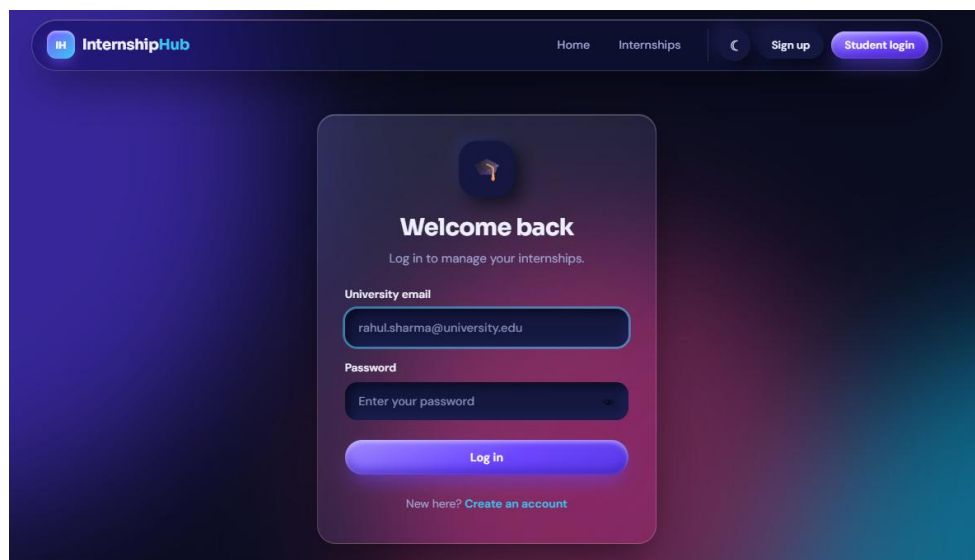


Figure 4. Student Login Screen

Student Registration:

The registration screen allows a new student to create an account by providing the required personal and academic information.

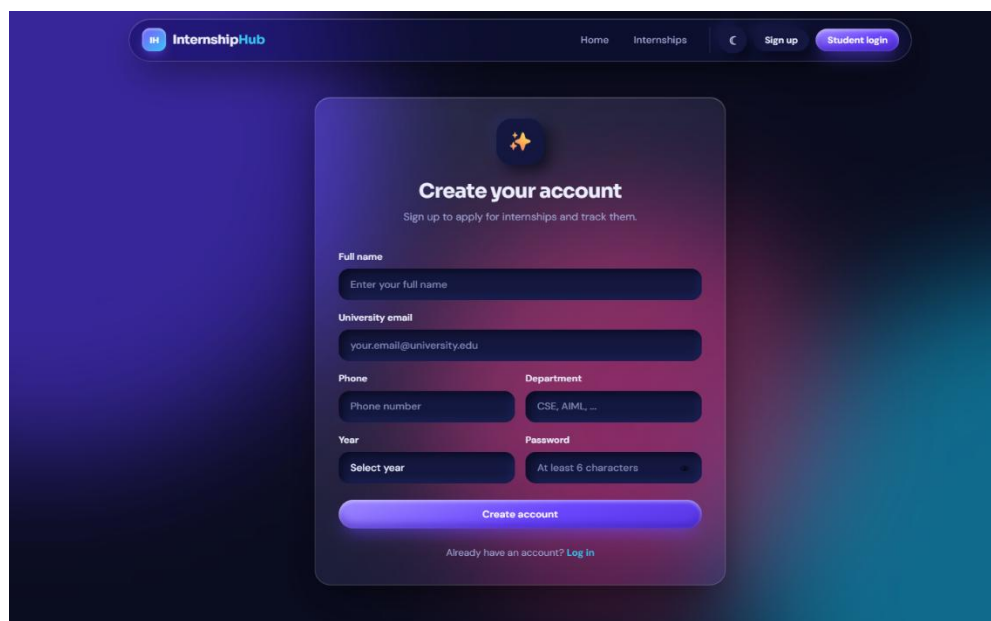


Figure 5. Student Registration Screen

Student Dashboard:

After successful login, the dashboard displays student information and application-related details.

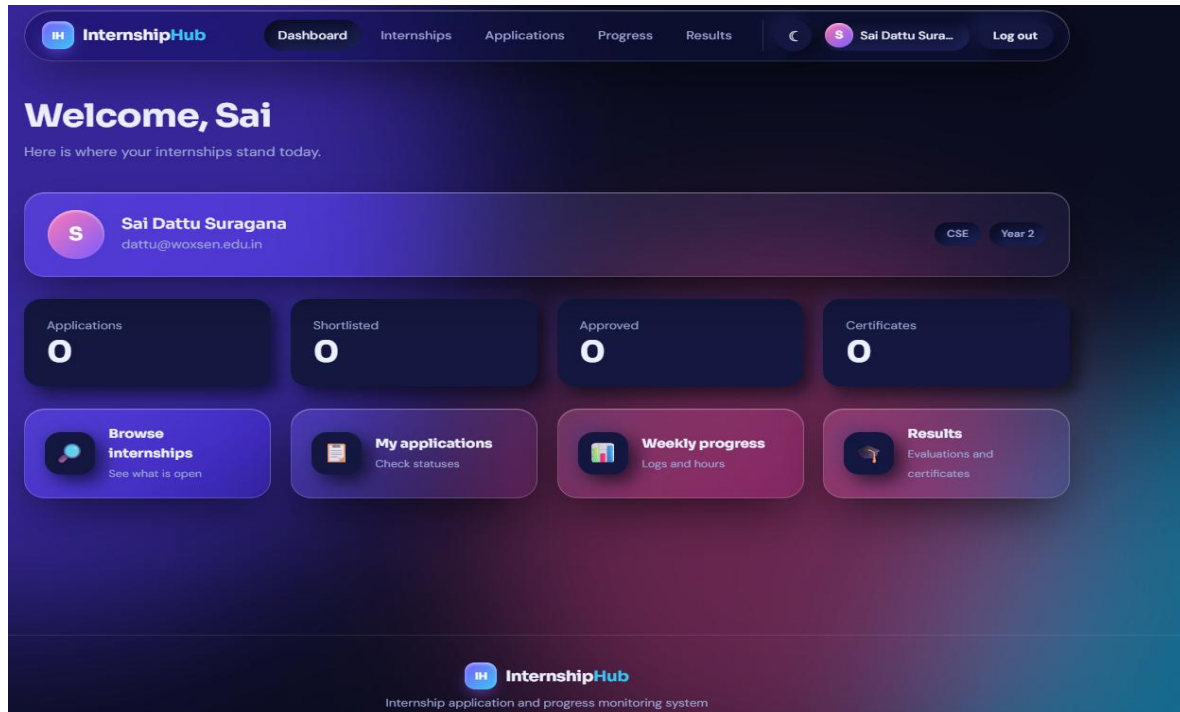


Figure 6. Student Dashboard

Internship Opportunities:

The opportunities page displays available internships along with organization, location, internship title, and available slots.

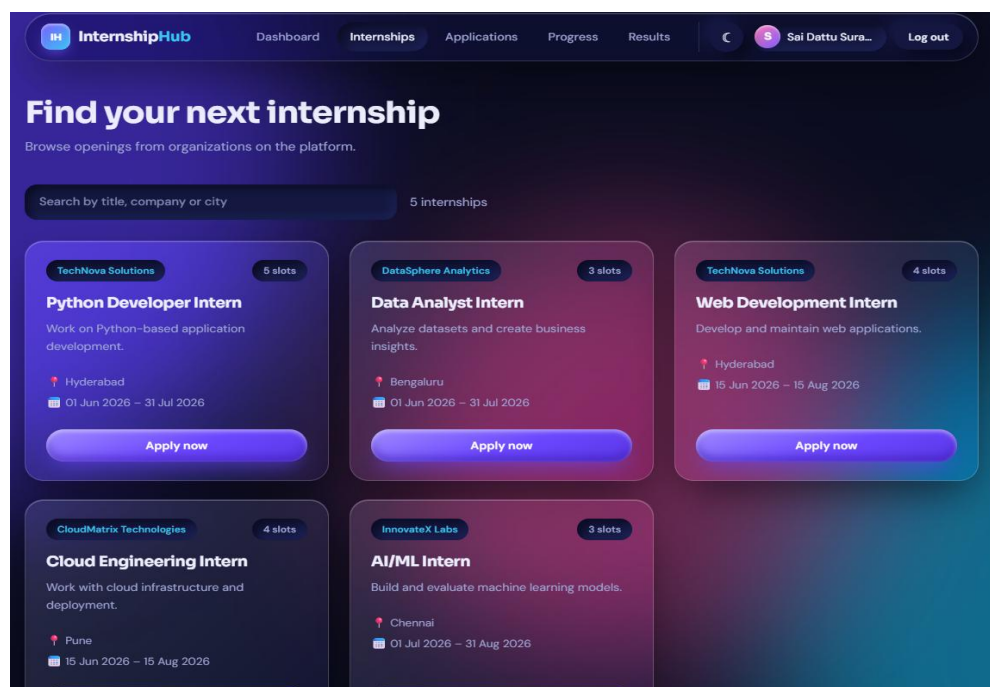


Figure 7. Internship Opportunities Screen

My Applications:

The applications page allows students to view their submitted internship applications and track their current status.

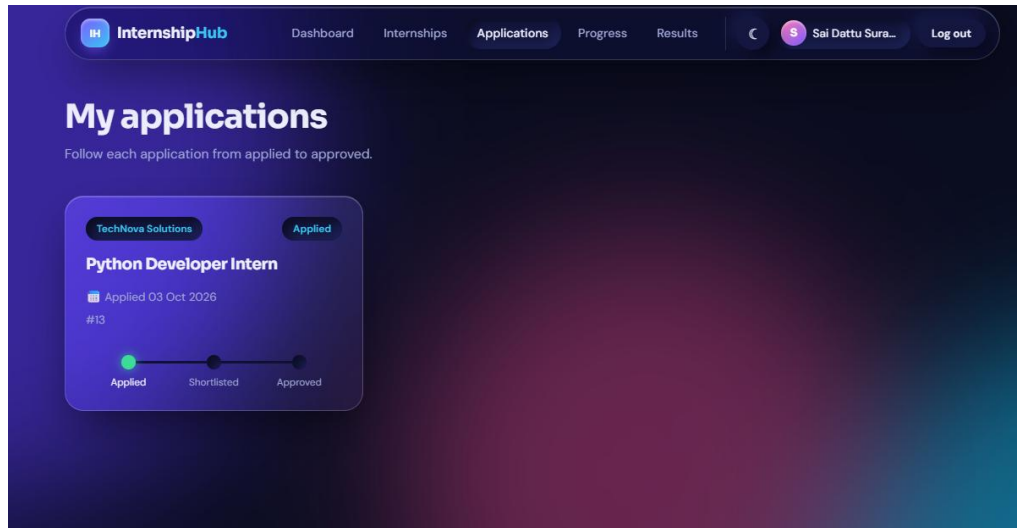


Figure 8. My Applications Screen

Weekly Progress:

The progress page provides access to weekly internship progress records associated with the student's assignment.

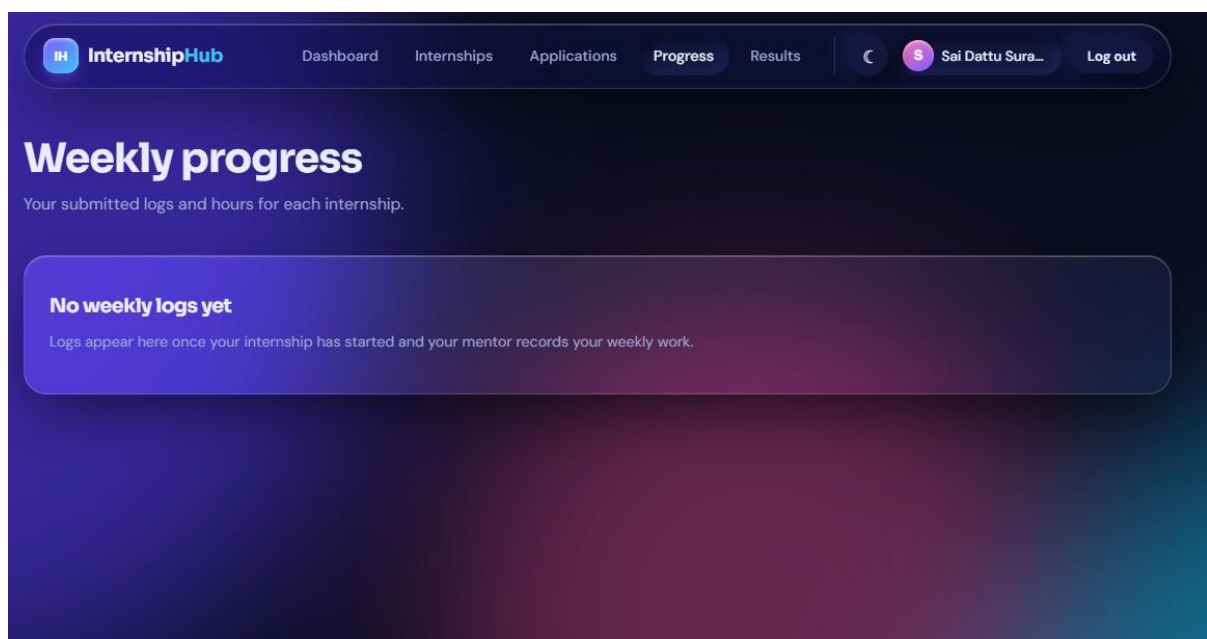


Figure 9. Weekly Progress Screen

Results:

The results page displays internship evaluation, certificate, and feedback information associated with the student's internship assignment.

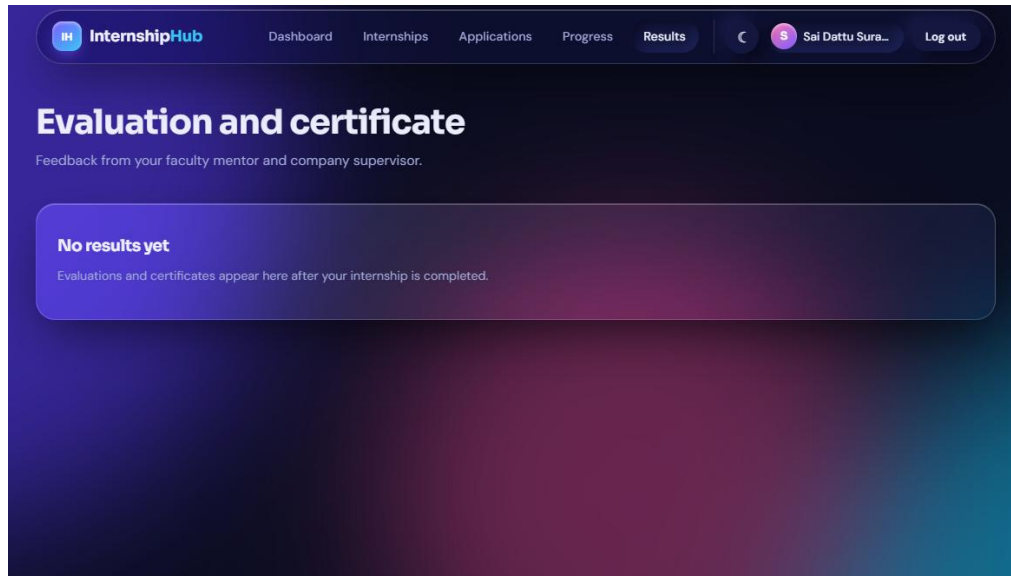


Figure 10. Results Screen

Admin Dashboard:

The Admin Dashboard provides the administrator with an overview of the internship management system. It displays application-related information and provides access to administrative operations such as internship management, application review, and internship assignment.

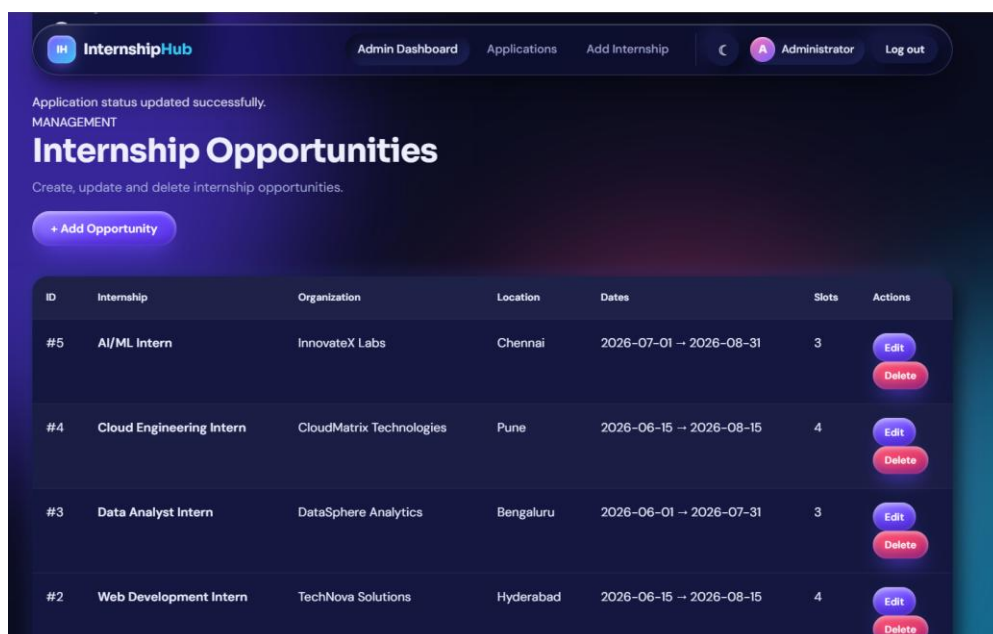
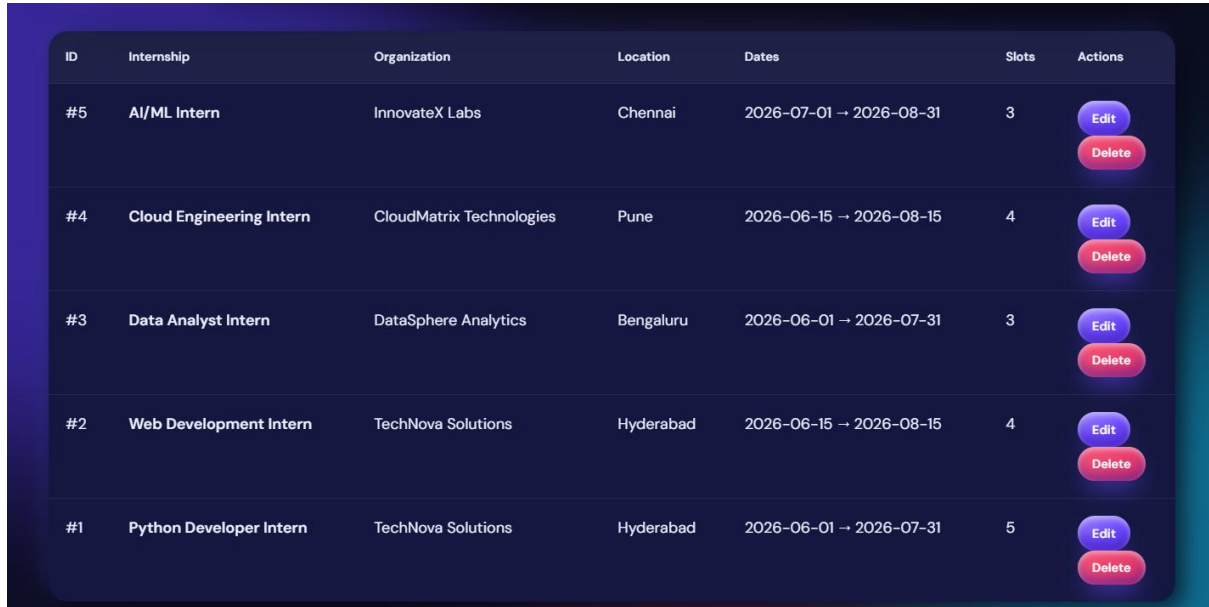


Figure 11. Admin Dashboard

Internship Management:

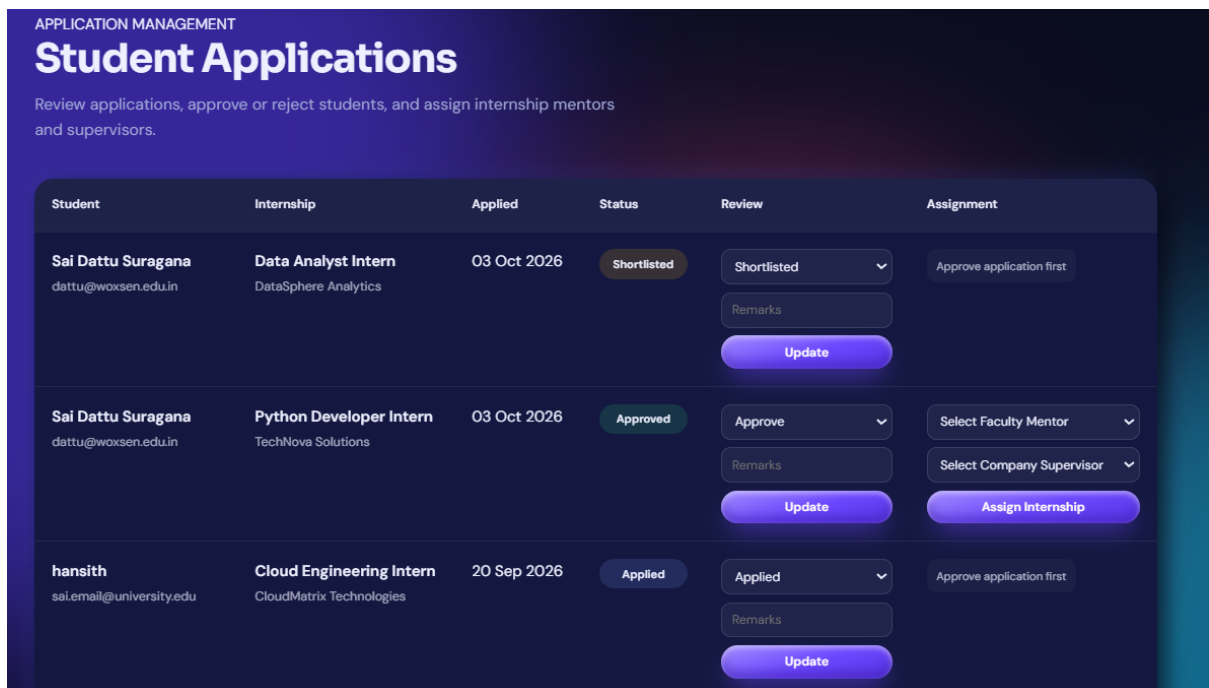
The Internship Management section allows the administrator to create, view, update, and delete internship opportunities



ID	Internship	Organization	Location	Dates	Slots	Actions
#5	AI/ML Intern	InnovateX Labs	Chennai	2026-07-01 → 2026-08-31	3	<button>Edit</button> <button>Delete</button>
#4	Cloud Engineering Intern	CloudMatrix Technologies	Pune	2026-06-15 → 2026-08-15	4	<button>Edit</button> <button>Delete</button>
#3	Data Analyst Intern	DataSphere Analytics	Bengaluru	2026-06-01 → 2026-07-31	3	<button>Edit</button> <button>Delete</button>
#2	Web Development Intern	TechNova Solutions	Hyderabad	2026-06-15 → 2026-08-15	4	<button>Edit</button> <button>Delete</button>
#1	Python Developer Intern	TechNova Solutions	Hyderabad	2026-06-01 → 2026-07-31	5	<button>Edit</button> <button>Delete</button>

Figure 12. Internship Management Screen

Student Application Review and Assignment:



APPLICATION MANAGEMENT						
Student Applications						
Review applications, approve or reject students, and assign internship mentors and supervisors.						
Student	Internship	Applied	Status	Review	Assignment	
Sai Dattu Suragana dattu@woxsen.edu.in	Data Analyst Intern DataSphere Analytics	03 Oct 2026	Shortlisted	Shortlisted Remarks <button>Update</button>	Approve application first	
Sai Dattu Suragana dattu@woxsen.edu.in	Python Developer Intern TechNova Solutions	03 Oct 2026	Approved	Approve Remarks <button>Update</button>	Select Faculty Mentor Select Company Supervisor	<button>Assign Internship</button>
hansith sailemail@university.edu	Cloud Engineering Intern CloudMatrix Technologies	20 Sep 2026	Applied	Applied Remarks <button>Update</button>	Approve application first	

Figure 13. Student Application Review and Assignment Screen

3.8 Testing and Results:

The Internship Application and Progress Monitoring System was tested to ensure that all the main parts functioned correctly and that the database operations worked as intended. Both good and bad test cases were used to verify how the application responds when data is validated and when database rules are applied.

Table 11. Testing and Results:

Test ID	Test Case	Expected Result	Status
T01	Student registration with valid details	Student account created successfully	Pass
T02	Registration using existing email	Registration rejected	Pass
T03	Student login with valid credentials	Student dashboard displayed	Pass
T04	Login with invalid password	Login rejected with error message	Pass
T05	View internship opportunities	Available internships displayed	Pass
T06	Submit internship application	Application stored with Applied status	Pass
T07	Apply for same internship again	Duplicate application prevented	Pass
T08	Admin approves application	Status updated to Approved	Pass
T09	Admin rejects application	Status updated to Rejected	Pass
T10	Assign mentor and supervisor	Internship assignment created successfully	Pass
T11	Assign supervisor from another organization	Assignment rejected	Pass
T12	Create, update and delete internship	CRUD operations performed successfully	Pass
T13	Enter invalid internship dates	Invalid data rejected	Pass
T14	Enter invalid evaluation/feedback values	Validation prevents invalid data	Pass

Testing Summary:

The tests confirmed that the main workflows, for students and administrators work as expected. Students can register, log in view internship opportunities, submit applications and track their application status.

Testing also confirmed that the Flask application connects properly with the MySQL database. This connection enables all components of the system to work smoothly and reliably.

Admin portal after students submitting the application to an intership:

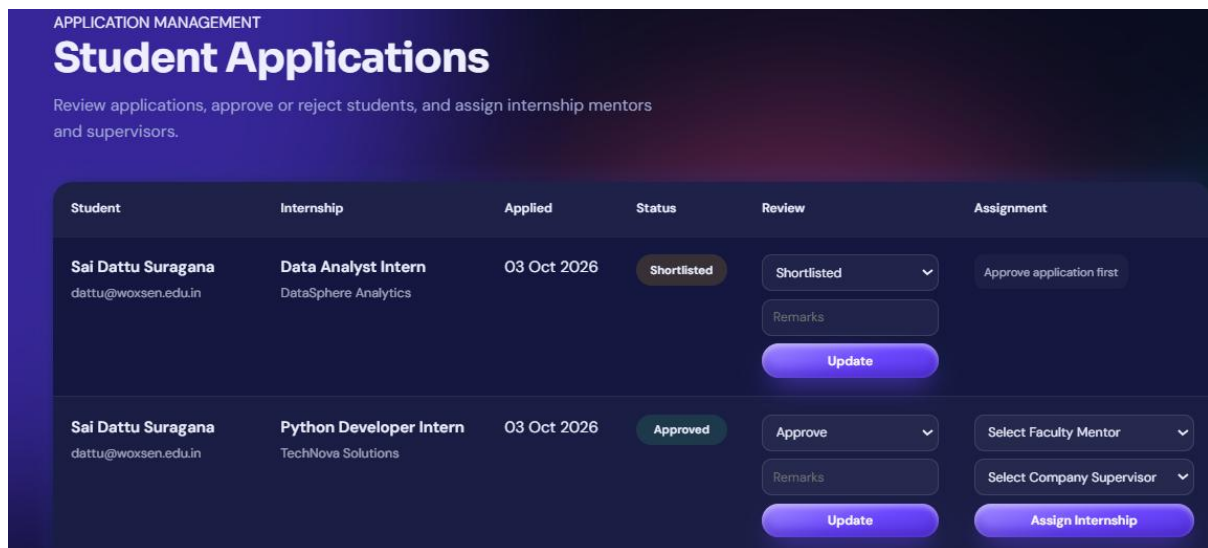


Figure 14: admin portal reviewing students' application

Student portal after admin reviews their application:

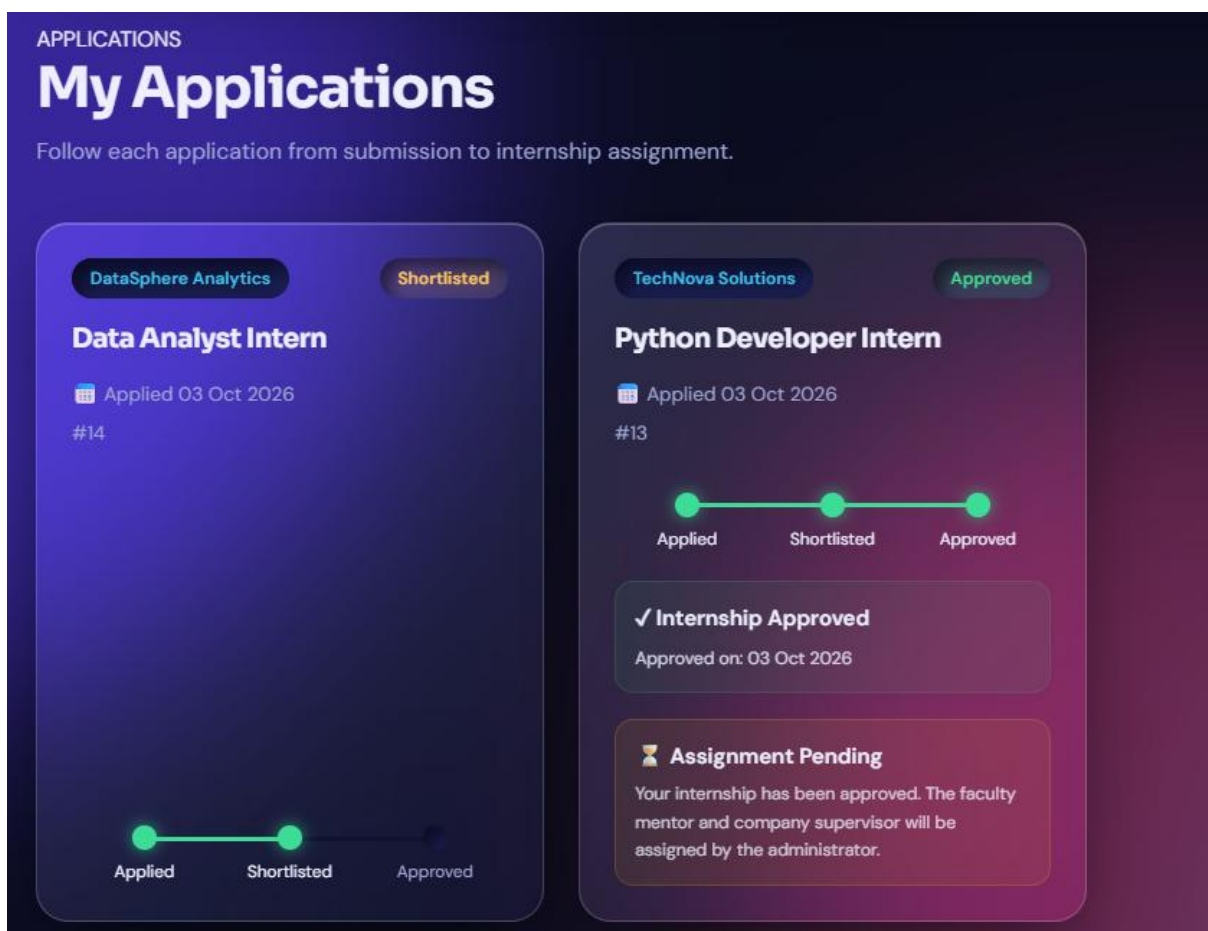


Figure 15: students portal application progress

CHAPTER 4: CONCLUSION, LIMITATIONS AND FUTURE ENHANCEMENTS

4.1 Conclusion:

The Internship Application and Progress Monitoring System also known as IAPMS provides a location to manage the entire student internship journey. The Internship Application and Progress Monitoring System stores student details organization data, internship opportunities, applications, approvals, assignments, weekly progress reports, evaluations, certificates and feedback all in one place.

The Internship Application and Progress Monitoring System was built with Python Flask and MySQL. The Internship Application and Progress Monitoring System allows students to register, sign in view internship opportunities, submit applications check application status and view internship results. The administrator can add internship opportunities, review student applications approve or reject them and assign faculty mentors and company supervisors.

The database of the Internship Application and Progress Monitoring System follows rules and is normalised to Third Normal Form. Primary keys, foreign keys, unique constraints check constraints and validation checks keep the data accurate. The Internship Application and Progress Monitoring System uses SQL queries, joins, summarising operations and views to retrieve and display internship data.

Overall the Internship Application and Progress Monitoring System demonstrates how a relational database and a web application can work together to organise and track internship applications and progress.

4.2 Limitations:

- The Internship Application and Progress Monitoring System relies on project-level credentials for administrator authentication.
- The Internship Application and Progress Monitoring System does not provide login screens for faculty mentors or company supervisors.
- The Internship Application and Progress Monitoring System does not yet send email or SMS notifications.

- The Internship Application and Progress Monitoring System does not allow uploading resumes or internship documents.
- The Internship Application and Progress Monitoring System has only been tested locally.
- The Internship Application and Progress Monitoring System offers analytics and limited dashboard views.

4.3 Future Enhancements:

- The Internship Application and Progress Monitoring System will add role-based login for administrators, faculty mentors and company supervisors.
- The Internship Application and Progress Monitoring System will send email alerts about application status, approvals and weekly log reminders.
- The Internship Application and Progress Monitoring System will allow users to upload resumes and internship documents.
- The Internship Application and Progress Monitoring System will provide analytics and performance dashboards for internships.
- The Internship Application and Progress Monitoring System will send reminders for pending applications and evaluations.
- The Internship Application and Progress Monitoring System will generate PDF and Excel reports.
- The Internship Application and Progress Monitoring System will move to the cloud so users can access it from anywhere.
- The Internship Application and Progress Monitoring System will enhance security, with environment variables and proper production authentication methods.

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Appendix A

COMPLETE DATABASE SQL SCRIPT

This appendix contains the complete SQL script used to create and initialize the **Internship Application and Progress Monitoring System** database. The script includes database creation, table definitions, primary keys, foreign keys, constraints, sample data, and database views.

The database is implemented using MySQL and contains the following 12 tables:

1. Student
2. Organization
3. Opportunity
4. Application
5. Approval
6. Faculty Mentor
7. Company Supervisor
8. Assignment
9. Weekly Log
10. Evaluation
11. Certificate
12. Feedback

The complete SQL script is provided to support database reproduction, testing, and verification of the implemented system.

A.1 Database and Table Creation:

[\[Click here to view the complete SQL script\]](#)

APPENDIX B

APPLICATION SOURCE CODE AND REPOSITORY REFERENCE

B.1 Application Source Code:

I built the source code for the Internship Application and Progress Monitoring System (IAPMS) using Python Flask, HTML, CSS, JavaScript and MySQL Connector/Python. The application includes features such as student authentication, internship opportunity management, application processing, administrative operations database connectivity, progress monitoring and result display.

The project is divided into parts: application logic, template files, static resources and database components. This structure helps keep the development environment organized and easy to manage.

B.2 Project Structure:

The major project structure is as follows:

```
Internship_Management_System/
```

```
|
```

```
|— backend/
```

```
|   |— app.py
```

```
|   |— database.py
```

```
|   |— test_connection.py
```

```
|   |
```

```
|   |— templates/
```

```
|       |— index.html
```

```
|       |— login.html
```

```
|       |— register.html
```

```
|       |— dashboard.html
```

```
|   |   └─ opportunities.html
|   |   └─ applications.html
|   |   └─ progress.html
|   |   └─ results.html
|   |   └─ admin.html
|   |
|   └─ static/
|       └─ css/
|           └─ style.css
|       └─ js/
|       └─ images/
|
└─ database/
    └─ internship_management.sql
```

All the source code, database SQL script, project files and supporting documentation will be stored in the project's GitHub repository. This way everything is kept in one place. Can be accessed easily.

GitHub Repository:

[\[Click here to view the complete project repository\]](#)

The GitHub repository contains the implementation. It allows others to review the source files, from the project report.

APPENDIX C

ADDITIONAL SQL QUERIES AND OUTPUTS

C.1 SQL Query Operations:

The following SQL operations were implemented and tested in the Internship Application and Progress Monitoring System to retrieve and analyze internship-related information.

S. No.	SQL Operation	Purpose
1	Student Applications	Display student applications with internship and organization details
2	Approved Internships	Retrieve applications with approved status
3	Assignment Details	Display assigned faculty mentors and company supervisors
4	Weekly Progress	Retrieve weekly internship activities and working hours
5	Evaluation Report	Display evaluation scores and remarks
6	Average Evaluation	Calculate average evaluation scores using AVG()
7	Applications by Organization	Count applications for each organization using COUNT()
8	Certificate Report	Display certificate details for assigned students
9	Internship Assignment View	Retrieve assignment information using the database view

C.2 Database View:

The project includes the following database view:

```
CREATE VIEW internship_assignment_report AS

SELECT

    ass.assignment_id,

    s.name AS student,

    s.department,

    org.name AS organization,

    op.title AS internship,

    fm.name AS faculty_mentor,

    cs.name AS company_supervisor,

    ass.assigned_date

FROM assignment ass

JOIN application ap

    ON ass.application_id = ap.application_id

JOIN student s

    ON ap.student_id = s.student_id

JOIN opportunity op

    ON ap.opportunity_id = op.opportunity_id

JOIN organization org

    ON op.organization_id = org.organization_id

JOIN faculty_mentor fm

    ON ass.mentor_id = fm.mentor_id

JOIN company_supervisor cs

    ON ass.supervisor_id = cs.supervisor_id;
```

C.3 Representative Query:

```
SELECT
    s.name AS student,
    o.title AS internship,
    org.name AS organization,
    a.application_date,
    a.status
FROM application a
JOIN student s
    ON a.student_id = s.student_id
JOIN opportunity o
    ON a.opportunity_id = o.opportunity_id
JOIN organization org
    ON o.organization_id = org.organization_id;
```

Purpose: This query combines information from the student, application, opportunity, and organization tables using joins to display internship application details.

C.4 Query Output

The output screenshots/tables of the important SQL queries can be included below this section if required by the faculty.

student	internship	organization	application_date	status
Rahul Sharma	Python Developer Intern	TechNova Solutions	2026-05-20	Approved
Sai Dattu Suragana	Python Developer Intern	TechNova Solutions	2026-10-03	Approved
Kiran Varma	Web Development Intern	TechNova Solutions	2026-05-24	Rejected
Sai Dattu Suragana	Web Development Intern	TechNova Solutions	2026-09-20	Applied
Priya Reddy	Data Analyst Intern	DataSphere Analytics	2026-05-21	Shortlisted
Rahul Sharma	Data Analyst Intern	DataSphere Analytics	2026-05-25	Shortlisted
Sai Dattu Suragana	Data Analyst Intern	DataSphere Analytics	2026-09-20	Applied
Sai Dattu Suragana	Data Analyst Intern	DataSphere Analytics	2026-10-03	Shortlisted
Arjun Kumar	Cloud Engineering Intern	CloudMatrix Technologies	2026-05-22	Applied
hansith	Cloud Engineering Intern	CloudMatrix Technologies	2026-09-20	Applied
Sneha Rao	AI/ML Intern	InnovateX Labs	2026-05-23	Approved
Priya Reddy	AI/ML Intern	InnovateX Labs	2026-05-26	Applied

Figure C.1: SQL Query Output for Student Internship Applications

APPENDIX D

ADDITIONAL TESTING EVIDENCE

D.1 Testing Evidence

The application was tested using representative positive and negative test cases covering student authentication, internship applications, administrative review, internship approval, mentor and company supervisor assignment, and internship opportunity management.

D.2 Test Case Summary:

Test ID	Test Scenario	Expected Result	Status
T01	Student registration	New student account created	Pass
T02	Duplicate email registration	Registration rejected	Pass
T03	Valid student login	Student successfully logged in	Pass
T04	Invalid password	Login rejected	Pass
T05	View internship opportunities	Available internships displayed	Pass
T06	Submit internship application	Application stored successfully	Pass
T07	Duplicate application	Duplicate application prevented	Pass
T08	Admin application approval	Application status updated to Approved	Pass
T09	Admin application rejection	Application status updated to Rejected	Pass
T10	Assign mentor and supervisor	Internship assignment created	Pass
T11	Invalid supervisor assignment	Assignment rejected	Pass
T12	Internship CRUD operations	Create, update, and delete operations performed	Pass

D.3 Additional Evidence:

The detailed application screens and functionality demonstrations are already presented in Chapter 3 – Application Development. Therefore, they are not repeated in this appendix.